EE/CS 148B HW4

Diffusion Models — Written Solutions

Alex Chung

Spring 2026

Contents

1 Part 1: DDPM — The Iterative Denoising Perspective 3

1.1 Problem 1.2: Variational Bound . . . 3
1.2 Problem 1.3: Tractable Loss Function . . . 3
1.3 Problem 1.4: Markov Process — Equation (4) . . . 4
1.4 Problem 1.5: Markov Process — Equations (6) and (7) . . . 5
1.5 Problem 1.6: KL Divergence and Equation (8) . . . 6
1.6 Problem 1.7: Parameterization and Equation (12) . . . 6
1.7 Problem 1.8: Coefficient Plot . . . 6

2 Part 2: Score Matching and Langevin Dynamics 6

2.1 Problem 2.2: Explicit Score Matching . . . 6
2.2 Problem 2.3: Implicit Score Matching . . . 7
2.3 Problem 2.4: Limitation of ISM . . . 8
2.4 Problem 2.5: Tweedie's Formula . . . 8
2.5 Problem 2.6: DSM Objective from Tweedie's Formula . . . 8
2.6 Problem 2.7: DSM vs. DDPM . . . 9
2.7 Problem 2.8: Langevin Dynamics Sampling . . . 9

3 Part 3: The SDE Perspective of Diffusion Models 9

3.1 Problem 3.2: VP-SDE Approximations . . . 9
3.2 Problem 3.3: Reverse VP-SDE . . . 10
3.3 Problem 3.4: Reverse VP-SDE Discretization . . . 10
3.4 Problem 3.5: Connecting SDE and DDPM . . . 10
3.5 Problem 3.6: Reverse SDE vs. Langevin Dynamics . . . 11
3.6 Problem 3.7: Conditional Diffusion and Classifier Guidance . . . 11

4 Part 4: Rectified Flow — Theory 11

4.1 Problem 4.1: Optimal Velocity Field . . . 11
4.2 Problem 4.2: Connection to Score Matching . . . 12
4.3 Problem 4.3: Straightness and Reflow . . . 12
4.4 Problem 4.4: Rectified Flow vs. DDPM . . . 13

5 Part 5: VP-SDE Written Questions 13

5.1 Problem 4.A.i: Drift and Diffusion Coefficients . . . 13
5.2 Problem 5.A.ii: $\beta(t)$ and $c(t)$ . . . 14

6 Part 5: VP-SDE Experimental Results 14
6.1 Problem 5.C.i: Dataset Visualization . . . 14
6.2 Problem 5.C.ii: Training Curves . . . 14
6.3 Problem 5.C.iii: EM Samples . . . 14
6.4 Problem 5.C.iv: PC Samples . . . 14
6.5 Problem 5.C.v: Qualitative Discussion . . . 14

7 Part 6: Rectified Flow Experimental Results 14
7.1 Problem 6.A: Forward Process and Loss . . . 14
7.2 Problem 6.B: Euler Sampler and Full Comparison . . . 15
7.3 Problem 6.C: One-Step Generation after Reflow . . . 15
7.4 Problem 6.D: Side-by-Side Qualitative Grid . . . 15

1 Part 1: DDPM — The Iterative Denoising Perspective

1.1 Problem 1.2: Variational Bound

Why minimize the variational bound on negative log likelihood.

We wish to learn a generative model p_θ by maximizing the log-likelihood $\mathbb{E}_{x_0 \sim q}[\log p_\theta(x_0)]$. Because the latent-variable marginal $p_\theta(x_0) = \int p_\theta(x_{0:T})\, dx_{1:T}$ involves an intractable integral over all latent trajectories, we derive a tractable surrogate via Jensen's inequality.

Proof of the chain of inequalities (Eq. 3 in DDPM).

For any forward process $q(x_{1:T} \mid x_0)$, multiply and divide inside the log to introduce q:

$$\begin{aligned}\log p_\theta(x_0) &= \log \int p_\theta(x_{0:T})\, dx_{1:T} \\ &= \log \int \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)}\, q(x_{1:T} \mid x_0)\, dx_{1:T} \\ &= \log \mathbb{E}_{q(x_{1:T}|x_0)}\left[\frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)}\right].\end{aligned}$$

Applying Jensen's inequality $\log \mathbb{E}[Z] \geq \mathbb{E}[\log Z]$ (since log is concave):

$$\log p_\theta(x_0) \geq \mathbb{E}_{q(x_{1:T}|x_0)}\left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)}\right].$$

Negating and taking the expectation over $x_0 \sim q(x_0)$ gives

$$\mathbb{E}_q[-\log p_\theta(x_0)] \leq \mathbb{E}_q\left[-\log \frac{p_\theta(x_{0:T})}{q(x_{1:T} \mid x_0)}\right] =: L. \tag{1}$$

Expanding the joint $p_\theta(x_{0:T}) = p(x_T)\prod_{t=1}^T p_\theta(x_{t-1} \mid x_t)$ and the Markov chain $q(x_{1:T} \mid x_0) = \prod_{t=1}^T q(x_t \mid x_{t-1})$:

$$L = \mathbb{E}_q\left[-\log p(x_T) - \sum_{t=1}^T \log \frac{p_\theta(x_{t-1} \mid x_t)}{q(x_t \mid x_{t-1})}\right],$$

which equals Eq. (3) of Ho et al. (2020).

Why we minimize L rather than maximize $\log p_\theta(x_0)$: L is a tractable upper bound on the NLL. Minimizing L pushes the ELBO up, thereby maximizing a guaranteed lower bound on $\log p_\theta(x_0)$.

1.2 Problem 1.3: Tractable Loss Function

We justify Equations (18), (20), (21), and (22) from Appendix A of Ho et al.

Eq. (18): Closed-form posterior $q(x_{t-1} \mid x_t, x_0)$.

By Bayes' rule and the Markov property:

$$q(x_{t-1} \mid x_t, x_0) \propto q(x_t \mid x_{t-1})\, q(x_{t-1} \mid x_0). \tag{2}$$

Using $q(x_t \mid x_{t-1}) = \mathcal{N}\big(x_t; \sqrt{\alpha_t}\, x_{t-1},\, \beta_t \mathbf{I}\big)$ and $q(x_{t-1} \mid x_0) = \mathcal{N}\big(x_{t-1}; \sqrt{\bar\alpha_{t-1}}\, x_0,\, (1-\bar\alpha_{t-1})\mathbf{I}\big)$, taking the log of the product and completing the square in x_{t-1}:

$$\begin{aligned}\log q(x_{t-1} \mid x_t, x_0) &\propto -\tfrac{1}{2}\left[\frac{(x_t - \sqrt{\alpha_t}\, x_{t-1})^2}{\beta_t} + \frac{(x_{t-1} - \sqrt{\bar\alpha_{t-1}}\, x_0)^2}{1-\bar\alpha_{t-1}}\right] \\ &= -\tfrac{1}{2}\left[\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar\alpha_{t-1}}\right) x_{t-1}^2 - 2\left(\frac{\sqrt{\alpha_t}}{\beta_t} x_t + \frac{\sqrt{\bar\alpha_{t-1}}}{1-\bar\alpha_{t-1}} x_0\right) x_{t-1} + C(x_t, x_0)\right].\end{aligned}$$

This is a Gaussian in x_{t-1} with precision $\sigma_t^{-2} = \frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}} = \frac{1-\bar{\alpha}_t}{\beta_t(1-\bar{\alpha}_{t-1})}$, so

$$\tilde{\beta}_t := \sigma_t^2 = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}, \qquad \tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\,x_t + \sqrt{\bar{\alpha}_{t-1}}\,\beta_t\,x_0}{1-\bar{\alpha}_t}. \tag{18}$$

Theorem used: Completion of the square for a Gaussian distribution; the product of two Gaussians is Gaussian (with precision equal to the sum of precisions).

Eq. (20): Reparameterised posterior mean.

From Eq. (6), $x_t = \sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon$ gives $x_0 = \big(x_t - \sqrt{1-\bar{\alpha}_t}\,\varepsilon\big)/\sqrt{\bar{\alpha}_t}$. Substituting into Eq. (18):

$$\tilde{\mu}_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}\beta_t\,(x_t - \sqrt{1-\bar{\alpha}_t}\,\varepsilon)/\sqrt{\bar{\alpha}_t}}{1-\bar{\alpha}_t}.$$

Using $\sqrt{\bar{\alpha}_{t-1}}/\sqrt{\bar{\alpha}_t} = 1/\sqrt{\alpha_t}$, the coefficient of x_t in the numerator is $\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1}) + \beta_t/\sqrt{\alpha_t} = [\alpha_t(1-\bar{\alpha}_{t-1}) + \beta_t]/\sqrt{\alpha_t} = (1-\bar{\alpha}_t)/\sqrt{\alpha_t}$ (using $\alpha_t + \beta_t = 1$ and $\bar{\alpha}_t = \alpha_t\bar{\alpha}_{t-1}$). Therefore:

$$\tilde{\mu}_t(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\varepsilon\right). \tag{20}$$

Equality used: Eq. (6), the reparameterisation $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon$.

Eq. (21): L_{t-1} as a mean squared error.

Setting $\Sigma_\theta(x_t, t) = \sigma_t^2\mathbf{I}$ (fixed variance) and noting that $q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t\mathbf{I})$ while $p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_\theta, \sigma_t^2\mathbf{I})$:

$$\begin{aligned} L_{t-1} &= D_{\mathrm{KL}}(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) \\ &= \frac{1}{2\sigma_t^2}\,\|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 + C', \end{aligned} \tag{21}$$

dropping terms that do not depend on θ. *Theorem used*: Closed-form KL divergence between two Gaussians with the same (diagonal) covariance, $D_{\mathrm{KL}}(\mathcal{N}(\mu_1, \Sigma) \,\|\, \mathcal{N}(\mu_2, \Sigma)) = \frac{1}{2}(\mu_1 - \mu_2)^\top\Sigma^{-1}(\mu_1 - \mu_2)$.

Eq. (22): Noise-parameterised simplified loss.

Define $\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\big(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\varepsilon_\theta(x_t, t)\big)$. Then

$$\begin{aligned} \|\tilde{\mu}_t - \mu_\theta\|^2 &= \left\|\frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon\right) - \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta\right)\right\|^2 \\ &= \frac{\beta_t^2}{\alpha_t(1-\bar{\alpha}_t)}\,\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2. \end{aligned}$$

Substituting into Eq. (21) with $\sigma_t^2 = \tilde{\beta}_t$ and taking the expectation:

$$L_{t-1} - C = \mathbb{E}_{x_0,\varepsilon}\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}\,\big\|\varepsilon - \varepsilon_\theta\big(\sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon,\, t\big)\big\|^2\right]. \tag{22}$$

1.3 Problem 1.4: Markov Process — Equation (4)

Claim: The forward process is a Markov chain: $q(x_{1:T} \mid x_0) = \prod_{t=1}^T q(x_t \mid x_{t-1})$.

Proof: By the chain rule of probability:

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^T q(x_t \mid x_{t-1}, \ldots, x_0). \tag{3}$$

The diffusion process is designed so that each step depends only on the previous state: $q(x_t \mid x_{t-1}, x_{t-2}, \ldots, x_0) = q(x_t \mid x_{t-1})$. This is the defining Markov property. Substituting:

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}). \tag{4}$$

Each transition is Gaussian: $q(x_t \mid x_{t-1}) = \mathcal{N}\big(x_t; \sqrt{1-\beta_t}\, x_{t-1},\, \beta_t \mathbf{I}\big)$.

1.4 Problem 1.5: Markov Process — Equations (6) and (7)

Eq. (6): Marginal $q(x_t \mid x_0)$.

We show by induction that $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$, where $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ and $\alpha_s = 1 - \beta_s$.

Base case ($t = 1$): From the transition $x_1 = \sqrt{\alpha_1} x_0 + \sqrt{\beta_1} \varepsilon_1$, we have $\bar{\alpha}_1 = \alpha_1$, which matches.

Inductive step: Assume $x_{t-1} = \sqrt{\bar{\alpha}_{t-1}} x_0 + \sqrt{1-\bar{\alpha}_{t-1}} \varepsilon_{t-1}$. Then:

$$\begin{aligned} x_t &= \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, \varepsilon_t \\ &= \sqrt{\alpha_t \bar{\alpha}_{t-1}}\, x_0 + \sqrt{\alpha_t (1-\bar{\alpha}_{t-1})}\, \varepsilon_{t-1} + \sqrt{\beta_t}\, \varepsilon_t. \end{aligned}$$

The two noise terms are independent Gaussians. By the additivity of Gaussian variances: $\sqrt{\alpha_t(1-\bar{\alpha}_{t-1})}\, \varepsilon_{t-1} + \sqrt{\beta_t}\, \varepsilon_t \overset{d}{=} \sqrt{\alpha_t(1-\bar{\alpha}_{t-1}) + \beta_t}\, \varepsilon = \sqrt{1-\bar{\alpha}_t}\, \varepsilon$, where the last step uses $\bar{\alpha}_t = \alpha_t \bar{\alpha}_{t-1}$ and $\alpha_t + \beta_t = 1$:

$$\alpha_t(1-\bar{\alpha}_{t-1}) + \beta_t = \alpha_t - \alpha_t \bar{\alpha}_{t-1} + 1 - \alpha_t = 1 - \bar{\alpha}_t.$$

Therefore $q(x_t \mid x_0) = \mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\, x_0,\, (1-\bar{\alpha}_t)\mathbf{I}\big)$.

Eq. (7): Posterior $q(x_{t-1} \mid x_t, x_0)$.

The posterior is derived via Bayes' rule using the joint distribution of (x_{t-1}, x_t) given x_0. Using the Gaussian conditioning formula:

Joint:

$$\begin{pmatrix} x_{t-1} \\ x_t \end{pmatrix} \Bigg| x_0 \sim \mathcal{N}\left(\begin{pmatrix} \sqrt{\bar{\alpha}_{t-1}} x_0 \\ \sqrt{\bar{\alpha}_t} x_0 \end{pmatrix}, \begin{pmatrix} (1-\bar{\alpha}_{t-1})\mathbf{I} & \sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{I} \\ \sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{I} & (1-\bar{\alpha}_t)\mathbf{I} \end{pmatrix} \right).$$

(The cross-covariance $\mathrm{Cov}(x_{t-1}, x_t \mid x_0) = \sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{I}$ follows from $x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_t$ and $\mathrm{Var}(x_{t-1} \mid x_0) = (1-\bar{\alpha}_{t-1})\mathbf{I}$.)

By the Gaussian conditioning formula:

$$\begin{aligned} x_{t-1} \mid x_t, x_0 &\sim \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t \mathbf{I}), \\ \tilde{\mu}_t &= \sqrt{\bar{\alpha}_{t-1}} x_0 + \sqrt{\alpha_t}(1-\bar{\alpha}_{t-1}) \cdot \frac{x_t - \sqrt{\bar{\alpha}_t} x_0}{1-\bar{\alpha}_t} \\ &= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1}) x_t + \sqrt{\bar{\alpha}_{t-1}} \beta_t x_0}{1-\bar{\alpha}_t}, \\ \tilde{\beta}_t &= (1-\bar{\alpha}_{t-1}) - \frac{\alpha_t (1-\bar{\alpha}_{t-1})^2}{1-\bar{\alpha}_t} = \frac{\beta_t (1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}. \end{aligned}$$

1.5 Problem 1.6: KL Divergence and Equation (8)

Setting $\Sigma_\theta(x_t, t) = \beta_t \mathbf{I}$ and using the closed-form KL formula for two k-dimensional Gaussians $p = \mathcal{N}(\mu_p, \Sigma_p)$ and $q = \mathcal{N}(\mu_q, \Sigma_q)$ with $\Sigma_p = \Sigma_q = \beta_t \mathbf{I}$:

$$D_{\mathrm{KL}}(p \,\|\, q) = \frac{1}{2}\left[\log \frac{|\Sigma_q|}{|\Sigma_p|} - k + \mathrm{tr}(\Sigma_q^{-1}\Sigma_p) + (\mu_q - \mu_p)^\top \Sigma_q^{-1}(\mu_q - \mu_p)\right].$$

When both distributions have the same covariance $\beta_t \mathbf{I}$, the first three terms cancel (log ratio is 0, trace term is k, $-k$ cancels), leaving:

$$L_{t-1} = D_{\mathrm{KL}}(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)) = \mathbb{E}_q\left[\frac{1}{2\beta_t}\left\|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\right\|^2\right] + C. \qquad (8)$$

Here C collects terms that do not depend on θ (e.g., constants from the log-determinant ratio and trace terms when the two covariances differ).

1.6 Problem 1.7: Parameterization and Equation (12)

Substituting the parameterization $\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\big(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t, t)\big)$ into Eq. (10) (which is Eq. (8) with $\sigma_t^2 = \tilde{\beta}_t$):

$$\begin{aligned}\|\tilde{\mu}_t - \mu_\theta\|^2 &= \left\|\frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon\right) - \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta\right)\right\|^2 \\ &= \frac{\beta_t^2}{\alpha_t(1-\bar{\alpha}_t)}\left\|\varepsilon - \varepsilon_\theta(x_t, t)\right\|^2.\end{aligned}$$

Substituting into Eq. (10) with $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon$:

$$L_{t-1} - C = \mathbb{E}_{x_0,\varepsilon}\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}\left\|\varepsilon - \varepsilon_\theta\big(\sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon,\, t\big)\right\|^2\right]. \qquad (12)$$

1.7 Problem 1.8: Coefficient Plot

The coefficient $\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}$ (with $\sigma_t^2 = \beta_t$) is plotted in Figure 1 using the linear schedule $\beta_t \in [10^{-4}, 0.02]$, $T = 1000$.

As noted in Section 3.4 of Ho et al., the simplified objective drops these time-dependent weights, which empirically leads to better sample quality by focusing training effort on the harder, high-noise timesteps.

2 Part 2: Score Matching and Langevin Dynamics

2.1 Problem 2.2: Explicit Score Matching

The explicit score matching (ESM) objective is:

$$\mathcal{J}_{\mathrm{ESM}}(\theta) = \mathbb{E}_{p(x)}\left[\tfrac{1}{2}\left\|s_\theta(x) - \nabla_x \log p(x)\right\|^2\right].$$

This is intractable in diffusion model applications for two reasons:

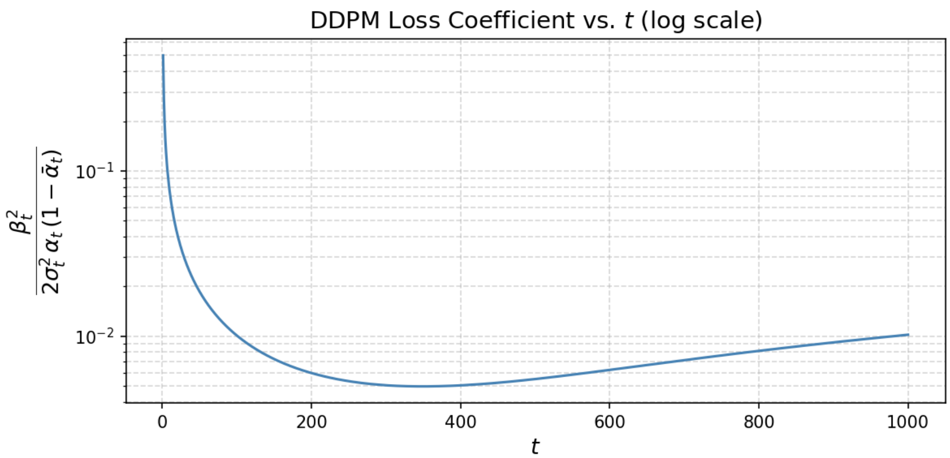

Figure 1: Loss coefficient $\beta_t^2/(2\sigma_t^2\alpha_t(1-\bar{\alpha}_t))$ vs. t (log y-axis). The coefficient is much larger for small t (low-noise, fine detail) and very small for large t (high-noise, coarse structure). The simplified objective (which drops these weights) is thus equivalent to down-weighting the fine-detail denoising terms.

1. **Unknown data score**: The true score $\nabla_x \log p(x)$ of the data distribution $p(x)$ is not available — if it were, we would not need to learn it.
2. **Complex marginals**: In diffusion models, $p(x)$ at intermediate noise levels is the marginal $p_t(x) = \int q(x \mid x_0) p_{\text{data}}(x_0)\, dx_0$, whose score is also unavailable in closed form for general data distributions.

2.2 Problem 2.3: Implicit Score Matching

Claim: $\mathcal{J}_{\text{ESM}}(\theta) = \mathcal{J}_{\text{ISM}}(\theta) + C$ where C does not depend on θ and $\mathcal{J}_{\text{ISM}}(\theta) = \mathbb{E}_{p(x)}\big[\operatorname{tr}(\nabla_x s_\theta(x)) + \frac{1}{2}\|s_\theta(x)\|^2\big]$.

Proof: Expanding the squared norm in ESM:

$$\mathcal{J}_{\text{ESM}}(\theta) = \mathbb{E}_p\left[\frac{1}{2}\|s_\theta\|^2 - s_\theta(x)^\top \nabla_x \log p(x) + \frac{1}{2}\|\nabla_x \log p\|^2\right].$$

The last term is a constant C independent of θ. It remains to show

$$\mathbb{E}_p\left[-s_\theta(x)^\top \nabla_x \log p(x)\right] = \mathbb{E}_p[\operatorname{tr}(\nabla_x s_\theta(x))].$$

Rewrite $\nabla_x \log p(x) = \frac{\nabla_x p(x)}{p(x)}$:

$$\mathbb{E}_p\left[-s_\theta(x)^\top \nabla_x \log p(x)\right] = -\int s_\theta(x)^\top \nabla_x p(x)\, dx.$$

Apply vector integration by parts (Green's first identity), using $p(x) \to 0$ as $\|x\| \to \infty$:

$$-\int s_\theta(x)^\top \nabla_x p(x)\, dx = \int p(x) \operatorname{div}(s_\theta(x))\, dx = \mathbb{E}_p[\operatorname{tr}(\nabla_x s_\theta(x))].$$

(The boundary term vanishes because p vanishes at infinity.) Therefore $\mathcal{J}_{\text{ESM}} = \mathcal{J}_{\text{ISM}} + C$. $\square$

2.3 Problem 2.4: Limitation of ISM

Despite being tractable, ISM requires computing $\mathrm{tr}(\nabla_x s_\theta(x))$ — the trace of the Jacobian of the network with respect to its input. For a network with d-dimensional input, this requires d backward passes (one per input dimension) or a Hutchinson estimator, making the per-step cost $O(d)$ times a standard gradient step. For high-dimensional inputs such as images ($d = 28 \times 28 = 784$ or larger), this is prohibitively expensive. Furthermore, estimating $\mathrm{tr}(\nabla_x s_\theta)$ via Hutchinson introduces additional variance.

2.4 Problem 2.5: Tweedie's Formula

Claim: Given $\tilde{x} = x + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$, $x \sim p(x)$:

$$\mathbb{E}[x \mid \tilde{x}] = \tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}).$$

Proof: Using the hint steps:
Step 1: We need $\mathbb{E}[x \mid \tilde{x}]$. Since $\tilde{x} = \mathbb{E}[\tilde{x} \mid \tilde{x}]$:

$$\mathbb{E}[x \mid \tilde{x}] = \tilde{x} - \mathbb{E}[\tilde{x} - x \mid \tilde{x}] = \tilde{x} - \mathbb{E}[\sigma\varepsilon \mid \tilde{x}].$$

Step 2: Write the conditional expectation as an integral:

$$\mathbb{E}[\sigma\varepsilon \mid \tilde{x}] = \frac{\int \sigma\varepsilon\, p(\tilde{x} \mid x)\, p(x)\, dx}{p_\sigma(\tilde{x})} = \frac{\int (\tilde{x} - x)\, p(\tilde{x} \mid x)\, p(x)\, dx}{p_\sigma(\tilde{x})}.$$

Step 3: Note $p(\tilde{x} \mid x) = \mathcal{N}(\tilde{x}; x, \sigma^2\mathbf{I})$, so: $\frac{\tilde{x}-x}{\sigma^2}\, p(\tilde{x} \mid x) = -\nabla_{\tilde{x}} p(\tilde{x} \mid x)$. Applying Bayes' rule:

$$\int (\tilde{x} - x)\, p(\tilde{x} \mid x)\, p(x)\, dx = -\sigma^2 \int \nabla_{\tilde{x}} p(\tilde{x} \mid x)\, p(x)\, dx = -\sigma^2\, \nabla_{\tilde{x}} p_\sigma(\tilde{x}).$$

Step 4: Therefore:

$$\mathbb{E}[\sigma\varepsilon \mid \tilde{x}] = \frac{-\sigma^2\, \nabla_{\tilde{x}} p_\sigma(\tilde{x})}{p_\sigma(\tilde{x})} = -\sigma^2\, \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}).$$

Conclusion:

$$\mathbb{E}[x \mid \tilde{x}] = \tilde{x} + \sigma^2\, \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}). \qquad \square$$

2.5 Problem 2.6: DSM Objective from Tweedie's Formula

Training objective from Tweedie: Tweedie tells us $\mathbb{E}[x \mid \tilde{x}] = \tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$. We can therefore train $s_\theta(\tilde{x})$ to approximate the score by regressing onto the known quantity $(\mathbb{E}[x \mid \tilde{x}] - \tilde{x})/\sigma^2 = \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$. Since the conditional expectation is not directly accessible, we instead minimize:

$$\mathcal{J}_{\mathrm{train}}(\theta; \sigma) = \mathbb{E}_{p(x)} \mathbb{E}_{p(\tilde{x}|x)} \left[\tfrac{1}{2} \left\| s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log p(\tilde{x} \mid x) \right\|^2 \right],$$

which is the DSM objective $\mathcal{J}_{\mathrm{DSM}}(\theta; \sigma) = \mathbb{E}_{p(x)} \mathbb{E}_{p(\tilde{x}|x)} \big[\tfrac{1}{2} \left\| s_\theta(\tilde{x}) - \nabla_{\tilde{x}} \log p(\tilde{x} \mid x) \right\|^2 \big]$.

Proof of equivalence with ESM: Expand:

$$\mathcal{J}_{\mathrm{DSM}} = \mathbb{E}\Big[\tfrac{1}{2} \left\| s_\theta(\tilde{x}) \right\|^2 - s_\theta(\tilde{x})^\top \nabla_{\tilde{x}} \log p(\tilde{x} \mid x) + \tfrac{1}{2} \left\| \nabla_{\tilde{x}} \log p(\tilde{x} \mid x) \right\|^2 \Big].$$

The last term is C independent of θ. For the cross term, noting $\nabla_{\tilde{x}} \log p(\tilde{x} \mid x) = -(\tilde{x}-x)/\sigma^2$:

$$\begin{aligned}
\mathbb{E}_{\tilde{x},x}\Big[-s_\theta(\tilde{x})^\top \nabla_{\tilde{x}} \log p(\tilde{x} \mid x)\Big] &= \mathbb{E}_{\tilde{x}}\left[s_\theta(\tilde{x})^\top \underbrace{\mathbb{E}_x\left[\frac{\tilde{x}-x}{\sigma^2}\middle|\tilde{x}\right]}_{=-\nabla_{\tilde{x}} \log p_\sigma(\tilde{x})\ \text{(from score identity)}}\right] \\
&= -\mathbb{E}_{\tilde{x}}\Big[s_\theta(\tilde{x})^\top \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})\Big] \\
&= \mathbb{E}_{\tilde{x}}[\operatorname{tr}(\nabla_{\tilde{x}} s_\theta(\tilde{x}))] \quad \text{(by integration by parts, as in ISM).}
\end{aligned}$$

Thus $\mathcal{J}_{\text{DSM}} = \mathcal{J}_{\text{ISM}} + C = \mathcal{J}_{\text{ESM}} + C$. □

2.6 Problem 2.7: DSM vs. DDPM

Setting $x = \alpha_t x_0$, $\tilde{x} = x_t$, $\sigma = \sqrt{1-\bar{\alpha}_t}$, the DSM objective at noise level $\sigma = \sqrt{1-\bar{\alpha}_t}$ becomes:

$$\mathcal{J}_{\text{DSM}} = \mathbb{E}\Big[\tfrac{1}{2}\left\|s_\theta(x_t) - \nabla_{x_t} \log q(x_t \mid x_0)\right\|^2\Big].$$

Since $\nabla_{x_t} \log q(x_t \mid x_0) = -\varepsilon/\sqrt{1-\bar{\alpha}_t}$, we have $s_\theta^*(x_t) = -\varepsilon^*/\sqrt{1-\bar{\alpha}_t}$. The simplified DDPM objective at time t is: $\|\varepsilon - \varepsilon_\theta(x_t,t)\|^2$, whose minimizer satisfies $\varepsilon_\theta^*(x_t,t) = \varepsilon^*$. Therefore:

$$s_{\theta^*}(x_t) = -\frac{\varepsilon_\theta^*(x_t,t)}{\sqrt{1-\bar{\alpha}_t}}, \quad \text{i.e.,} \quad s_{\theta^*} = c(\sigma)\,\varepsilon_\theta^*, \quad c(\sigma) = -\frac{1}{\sigma} = -\frac{1}{\sqrt{1-\bar{\alpha}_t}}.$$

The two objectives share the same optimizer, differing by the constant scalar $c(\sigma) = -1/\sigma$.

2.7 Problem 2.8: Langevin Dynamics Sampling

The overdamped Langevin SDE $dx_t = \nabla \log p_\sigma(x_t)\,dt + \sqrt{2}\,dB_t$ admits $p_\sigma(x)$ as its stationary distribution. For small σ, $p_\sigma \approx p$.

Algorithm:

1. Train a score model $s_\theta(\tilde{x}) \approx \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$ via DSM with a small noise level σ.
2. Initialize $x_0 \sim \mathcal{N}(0, \mathbf{I})$ (or any distribution).
3. For $k = 0, 1, \ldots, K-1$ (with step size $\eta > 0$):

$$x_{k+1} = x_k + \eta\, s_\theta(x_k) + \sqrt{2\eta}\, z_k, \quad z_k \sim \mathcal{N}(0, \mathbf{I}).$$

4. Return x_K as an approximate sample from $p(x)$.

As $K \to \infty$ (and $\eta \to 0$ appropriately), $x_K \sim p_\sigma \approx p$.

3 Part 3: The SDE Perspective of Diffusion Models

3.1 Problem 3.2: VP-SDE Approximations

In Appendix B of Song et al. (2021), the DDPM forward process with $x_{i+1} = \sqrt{1-\beta_i}x_i + \sqrt{\beta_i}z_i$ is shown to approximate the VP-SDE $dx = -\frac{1}{2}\beta(t)x\,dt + \sqrt{\beta(t)}\,dB_t$ via two approximations as $\Delta t \ll 1$:

1. $\sqrt{1-\beta_i} \approx 1 - \frac{1}{2}\beta_i$. This holds since $\sqrt{1-\beta} \approx 1 - \beta/2$ for $\beta \ll 1$ (first-order Taylor expansion of $\sqrt{1-\beta}$ around $\beta = 0$). The error is $O(\beta^2) = O(\Delta t^2)$, which vanishes as $\Delta t \to 0$.

2. $\sqrt{\beta_i} \approx \sqrt{\beta(t)\Delta t}$. With $\beta_i = \bar{\beta}_i \Delta t$ (the discrete noise level set proportional to Δt so that the continuous limit is finite), and $\bar{\beta}_i \to \beta(t)$ as $\Delta t \to 0$, we get $\sqrt{\beta_i} = \sqrt{\bar{\beta}_i \Delta t} \approx \sqrt{\beta(t)\Delta t}$. This is the standard Itô SDE discretization term $g\sqrt{\Delta t}$.

Under these two approximations the discrete update $x_{i+1} = \sqrt{1-\beta_i}x_i + \sqrt{\beta_i}z_i$ becomes $x_{t+\Delta t} = x_t - \frac{1}{2}\beta(t)x_t\,\Delta t + \sqrt{\beta(t)\Delta t}\,z$, which is exactly the Euler-Maruyama discretization of the VP-SDE.

3.2 Problem 3.3: Reverse VP-SDE

The general formula for the reverse SDE of $dx = f(x,t)\,dt + g(t)\,dB_t$ is (Anderson, 1982; Song et al. Eq. (6)):

$$dx = \left[f(x,t) - g(t)^2\,\nabla_x \log p_t(x)\right]dt + g(t)\,d\bar{B}_t,$$

where $d\bar{B}_t$ is the time-reversed Brownian motion. For the VP-SDE with $f(x,t) = -\frac{1}{2}\beta(t)x$ and $g(t) = \sqrt{\beta(t)}$:

$$\boxed{dx = \left[-\tfrac{1}{2}\beta(t)\,x - \beta(t)\,\nabla_x \log p_t(x)\right]dt + \sqrt{\beta(t)}\,d\bar{B}_t.} \tag{5}$$

3.3 Problem 3.4: Reverse VP-SDE Discretization

Step 1: Discretize Eq. (5).

With $\beta_i = \bar{\beta}_i\Delta t$ and indices running from high noise $(i+1)$ to low noise (i):

$$\begin{aligned} x_i &= x_{i+1} - \Delta t\left[-\tfrac{1}{2}\beta(t_{i+1})x_{i+1} - \beta(t_{i+1})s_\theta(x_{i+1}, i+1)\right] + \sqrt{\beta(t_{i+1})\Delta t}\,z_{i+1} \\ &= x_{i+1} + \tfrac{1}{2}\beta_{i+1}x_{i+1} + \beta_{i+1}s_\theta(x_{i+1}, i+1) + \sqrt{\beta_{i+1}}\,z_{i+1}. \end{aligned} \tag{6}$$

Step 2: Substitute s_θ. Keep $s_\theta(x_{i+1}, i+1)$.

Step 3: Rearrange. From Eq. (6):

$$x_i = (1 + \tfrac{1}{2}\beta_{i+1})\,x_{i+1} + \beta_{i+1}s_\theta + \sqrt{\beta_{i+1}}z_{i+1}.$$

Step 4: Follow Appendix E of Song21. For $\beta_{i+1} \to 0$: $1 + \frac{1}{2}\beta_{i+1} \approx \frac{1}{\sqrt{1-\beta_{i+1}}}$ (Taylor expansion). Rearranging the $x_{i+1} + \beta_{i+1}s_\theta$ part:

$$\boxed{x_i = \frac{1}{\sqrt{1-\beta_{i+1}}}\big(x_{i+1} + \beta_{i+1}s_\theta(x_{i+1}, i+1)\big) + \sqrt{\beta_{i+1}}\,z_{i+1}.} \tag{7}$$

3.4 Problem 3.5: Connecting SDE and DDPM

DDPM Algorithm 2 uses the update (with $\alpha_t = 1 - \beta_t$, $\sigma_t = \sqrt{\beta_t}$):

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\varepsilon_\theta(x_t, t)\right) + \sigma_t z.$$

The VP-SDE update above has coefficient β_{i+1} in front of s_θ, whereas DDPM has $\beta_t/\sqrt{1-\bar{\alpha}_t}$ in front of ε_θ, divided by $\sqrt{\alpha_t}$. These are consistent because (from Problem 2.7):

$$s_\theta(x_t, t) = -\frac{\varepsilon_\theta(x_t, t)}{\sigma(t)},$$

where $\sigma(t) = \sqrt{1 - \bar{\alpha}_t}$ is the standard deviation of the noise added to x_0. Substituting and noting $\beta_{i+1}/(1-\bar{\alpha}_t)^{1/2} \cdot (1-\bar{\alpha}_t)^{1/2}$ matches the DDPM factor, the two update rules agree up to the scalar $c(\sigma) = -1/\sigma$ from Problem 2.7.

3.5 Problem 3.6: Reverse SDE vs. Langevin Dynamics

Extra information in the reverse SDE. Langevin dynamics uses only the marginal score $\nabla_x \log p(x)$ to sample from $p(x)$. The reverse VP-SDE uses the *time-indexed* score $\nabla_x \log p_t(x)$, which encodes how the data distribution evolves under a structured noise process. Specifically, it knows that the process starts as $\mathcal{N}(0, \sigma_T^2 \mathbf{I})$ at $t = T$ and approaches p_{data} as $t \to 0$, giving a principled annealing schedule.

Reflection in the score matching objective (Song21 Eq. 7). The VP-SDE training objective is $\mathbb{E}_t[\lambda(t)\mathbb{E}_{x_0,x_t}[\|s_\theta(x_t,t) + \varepsilon/\sigma(t)\|^2]]$, which trains s_θ at *every noise level* t to match the time-conditioned score. Langevin dynamics would only train at a single (small) σ.

Reflection in the DDPM training objective (Ho et al. Eq. 14). DDPM trains $\varepsilon_\theta(x_t,t)$ at every discrete time step $t \in \{1, \ldots, T\}$, explicitly conditioning on t. The time index carries the information about the current noise level, enabling the model to guide the reverse trajectory accurately rather than blindly following the gradient of a single stationary distribution.

3.6 Problem 3.7: Conditional Diffusion and Classifier Guidance

The reverse SDE for the conditional distribution $p(x \mid y)$ is:

$$dx = \left[f(x,t) - g(t)^2 \, \nabla_x \log p_t(x \mid y)\right] dt + g(t)\, d\bar{B}_t.$$

By Bayes' rule: $\nabla_x \log p_t(x \mid y) = \nabla_x \log p_t(x) + \nabla_x \log p_t(y \mid x)$.

Classifier guidance intuition. Train the unconditional score model $s_\theta(x,t) \approx \nabla_x \log p_t(x)$ and a *noisy classifier* $p_\phi(y \mid x_t, t)$ separately. At sampling time, combine: $\nabla_x \log p_t(x \mid y) \approx s_\theta(x_t,t) + \nabla_{x_t} \log p_\phi(y \mid x_t, t)$. The classifier gradient steers the sample toward the desired class y while the unconditional score ensures image quality. Increasing the classifier gradient weight (guidance scale) trades diversity for class fidelity.

4 Part 4: Rectified Flow — Theory

4.1 Problem 4.1: Optimal Velocity Field

Claim: The minimizer of $\mathcal{L}_{\text{RF}}(\theta) = \int_0^1 \mathbb{E}\left[\|(X_1 - X_0) - v_\theta(X_t,t)\|^2\right] dt$ is $v^*(x,t) = \mathbb{E}[X_1 - X_0 \mid X_t = x]$.

Proof: For fixed t, expand $\mathbb{E}[\|(X_1 - X_0) - v_\theta(X_t,t)\|^2]$ by conditioning on X_t:

$$\mathbb{E}\left[\|(X_1 - X_0) - v_\theta(X_t,t)\|^2\right] = \mathbb{E}_{X_t}\left[\mathbb{E}\left[\|(X_1 - X_0) - v_\theta(X_t,t)\|^2 \;\middle|\; X_t\right]\right].$$

For each fixed $X_t = x$, minimize over $v_\theta(x,t)$:

$$\begin{aligned} \frac{\partial}{\partial v} \mathbb{E}\left[\|(X_1 - X_0) - v\|^2 \mid X_t = x\right] &= 2\big(v - \mathbb{E}[X_1 - X_0 \mid X_t = x]\big) = 0 \\ \implies \quad v^*(x,t) &= \mathbb{E}[X_1 - X_0 \mid X_t = x]. \qquad \square \end{aligned}$$

(This is the pointwise conditional mean: for each x and t, the velocity that minimizes the squared error is the conditional expectation of the target.)

4.2 Problem 4.2: Connection to Score Matching

Claim: If $X_0 \sim \mathcal{N}(0, \mathbf{I})$ and p_t is the marginal density of $X_t = (1-t)X_0 + tX_1$, then

$$v^*(x,t) = \frac{x - \mathbb{E}[X_0 \mid X_t = x]}{t}.$$

Derivation: From the interpolation $X_t = (1-t)X_0 + tX_1$: $X_1 = (X_t - (1-t)X_0)/t$, so

$$\begin{aligned} v^*(x,t) &= \mathbb{E}[X_1 - X_0 \mid X_t = x] = \mathbb{E}\left[\frac{X_t - (1-t)X_0}{t} - X_0 \,\middle|\, X_t = x\right] \\ &= \frac{x}{t} - \frac{1-t}{t}\mathbb{E}[X_0 \mid X_t = x] - \mathbb{E}[X_0 \mid X_t = x] = \frac{x - \mathbb{E}[X_0 \mid X_t = x]}{t}. \end{aligned}$$

Relating $\mathbb{E}[X_0 \mid X_t = x]$ to the score via Tweedie's formula:

Since $X_0 \sim \mathcal{N}(0, \mathbf{I})$ and $X_t = tX_1 + (1-t)X_0$, conditioning on $X_1 = x_1$: $X_t \mid X_1 = x_1 \sim \mathcal{N}(tx_1, (1-t)^2\mathbf{I})$. The marginal score satisfies:

$$\begin{aligned} \nabla_x \log p_t(x) &= \mathbb{E}_{X_1|X_t=x}[\nabla_x \log p(x \mid X_1)] = \mathbb{E}_{X_1|X_t=x}\left[-\frac{x - tX_1}{(1-t)^2}\right] \\ &= -\frac{x - t\,\mathbb{E}[X_1 \mid X_t = x]}{(1-t)^2}. \end{aligned}$$

Using the constraint $t\mathbb{E}[X_1 \mid X_t = x] = x - (1-t)\mathbb{E}[X_0 \mid X_t = x]$ (from $\mathbb{E}[X_t \mid X_t = x] = x$):

$$\nabla_x \log p_t(x) = -\frac{(1-t)\mathbb{E}[X_0 \mid X_t = x]}{(1-t)^2} = -\frac{\mathbb{E}[X_0 \mid X_t = x]}{1-t}.$$

Therefore:

$$\mathbb{E}[X_0 \mid X_t = x] = -(1-t)\,\nabla_x \log p_t(x),$$

and the optimal velocity is:

$$\boxed{v^*(x,t) = \frac{x + (1-t)\,\nabla_x \log p_t(x)}{t}.} \tag{8}$$

Discussion: This reveals a direct connection between rectified flow and score-based diffusion: the optimal rectified flow velocity at time t is a linear combination of x and the score $\nabla \log p_t(x)$. In particular, learning the velocity field v_θ is equivalent to learning the score (up to a time-dependent linear transform), so rectified flow and DDPM/VP-SDE are learning essentially the same function.

4.3 Problem 4.3: Straightness and Reflow

(1) Geometric meaning of S.

The straightness metric $S = 1 - \frac{\mathbb{E}[\int_0^1 \|v_\theta(X_t,t)\,dt - (X_1 - X_0)\|^2\,dt]}{\mathbb{E}[\|X_1 - X_0\|^2]}$ (where the numerator integral is $\mathbb{E}[\|\int_0^1 v_\theta dt - (X_1 - X_0)\|^2]$).

- $S = 1$: The numerator is zero, meaning $\int_0^1 v_\theta(X_t, t)\,dt = X_1 - X_0$ exactly for all samples. The ODE trajectory from X_0 to X_1 is a *straight line*, so one Euler step suffices.
- $S = 0$: The integral of the velocity equals zero contribution on average relative to the true displacement; the trajectory is maximally curved, requiring many steps.

(2) Why reflow produces straighter trajectories.

After training the initial v_θ, the ODE flow $\Phi^1(X_0)$ maps fresh noise to approximate data. The reflow procedure generates new pairs $(\hat{X}_0, \hat{X}_1)$ where $\hat{X}_0$ and $\hat{X}_1 = \Phi^1(\hat{X}_0)$ are *deterministically* coupled through the ODE. The optimal velocity for these paired samples is $\hat{X}_1 - \hat{X}_0$ pointwise (since the pairing is deterministic, the conditional expectation collapses). Retraining on these pairs thus learns a velocity that better aligns with the straight-line path from $\hat{X}_0$ to $\hat{X}_1$, reducing the curvature of the ODE. In other words, the initial v_θ already has implicit pairing between noise and data; reflow makes this pairing *explicit* so that the new velocity can match it with fewer steps.

(3) Computational cost of reflow.

Each round of reflow requires: (a) generating N pairs by running K-step Euler ODE at inference ($O(NK)$ model evaluations), and (b) retraining for several epochs. In total, this is $\approx 2\times$ the original training cost per round. The main reason to limit rounds is diminishing returns: each reflow round reduces the sampling steps needed roughly from K to K/r for some $r < 2$, but the cost grows linearly. Also, the generated $(\hat{X}_0, \hat{X}_1)$ pairs are approximate, and repeated reflow may compound approximation errors.

4.4 Problem 4.4: Rectified Flow vs. DDPM

(1) Training objective. DDPM regresses the added noise $\varepsilon \in \mathcal{N}(0, \mathbf{I})$: $\|\varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon,\, t)\|^2$. Rectified flow regresses the velocity $X_1 - X_0$: $\|(X_1 - X_0) - v_\theta(X_t, t)\|^2$. DDPM's target is pure Gaussian noise, while rectified flow's target is a data-dependent vector.

(2) Inference cost. DDPM requires $T \approx 1000$ reverse steps because each step only removes a small β_t fraction of noise and the SDE trajectory is stochastic and curved. Rectified flow can use far fewer steps because the ODE trajectory is (approximately) straight; after reflow it can use exactly one Euler step, since the learned velocity field is constant along straight-line paths.

(3) Trajectory geometry. DDPM's reverse process follows a stochastic curved SDE path, leading to large discretization error for large step sizes. Rectified flow integrates a deterministic ODE along approximately straight trajectories, so the Euler method has small discretization error even with one or a few steps.

(4) Likelihood evaluation. DDPM's ELBO gives a variational lower bound on $\log p(x)$, enabling tractable likelihood estimation. Rectified flow, as a continuous normalizing flow via the ODE, can in principle compute exact log-likelihoods using the change-of-variables formula (instantaneous change of variables), but in practice this is expensive (requires trace of the Jacobian). The standard rectified flow training does not provide a direct likelihood estimate.

5 Part 5: VP-SDE Written Questions

5.1 Problem 4.A.i: Drift and Diffusion Coefficients

For the VP-SDE $dx = f(x,t)\,dt + g(t)\,dB_t$:

$$f(x,t) = -\tfrac{1}{2}\beta(t)\,x, \tag{9}$$
$$g(t) = \sqrt{\beta(t)}, \tag{10}$$

where $\beta(t) = \beta_{\min} + (\beta_{\max} - \beta_{\min})\,t$ is the linear noise schedule (Eq. (32) of Song21).

5.2 Problem 5.A.ii: $\beta(t)$ and $c(t)$

$\beta(t)$: Linear schedule (Song21 Eq. 32):

$$\beta(t) = \beta_{\min} + (\beta_{\max} - \beta_{\min})\, t.$$

$c(t)$: The signal decay factor (Song21 Eq. 33):

$$c(t) = \exp\left(-\tfrac{1}{2}\int_0^t \beta(s)\, ds\right) = \exp\left(-\tfrac{1}{2}\left[\beta_{\min}\, t + \tfrac{1}{2}(\beta_{\max} - \beta_{\min})\, t^2\right]\right).$$

Consequently $\sigma(t) = \sqrt{1 - c(t)^2}$, and the marginal is $x_t = c(t)x_0 + \sigma(t)\varepsilon$, $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$.

6 Part 5: VP-SDE Experimental Results

6.1 Problem 5.C.i: Dataset Visualization

FashionMNIST contains 70,000 grayscale images (28×28 pixels, 1 channel) across 10 clothing classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot. Sample images are shown in Figure 2.

6.2 Problem 5.C.ii: Training Curves

6.3 Problem 5.C.iii: EM Samples

6.4 Problem 5.C.iv: PC Samples

6.5 Problem 5.C.v: Qualitative Discussion

The EM sampler produces recognizable FashionMNIST images but with some blurring at fine details. The PC sampler with more corrector steps tends to produce sharper and more coherent samples because the Langevin corrector steps reduce accumulated discretization error at each noise level. Comparing to the dataset (Figure 2), the model captures the global structure of each clothing item well (silhouettes, overall shape) but may struggle with fine textures and small details, which is consistent with the smaller loss coefficients at low t (Figure 1) causing less training signal on fine-detail denoising.

7 Part 6: Rectified Flow Experimental Results

7.1 Problem 6.A: Forward Process and Loss

The rectified flow forward process and training loss are implemented in `diffusion/rectflow.py`. The interpolation is $X_t = (1-t)X_0 + tX_1$ with $X_0 \sim \mathcal{N}(0, \mathbf{I})$, and the loss is MSE on the velocity $X_1 - X_0$.

The VP (DDPM) training loss and the rectified flow loss operate at different scales: DDPM regresses unit-norm Gaussian noise ε, while rectified flow regresses $X_1 - X_0$ which has roughly $\sqrt{2}\times$ larger norm on average (sum of two independent unit-norm Gaussians). Thus the RF loss values are approximately $2\times$ higher than the DDPM loss — they cannot be directly compared on the same scale.

Table 1: KID scores (mean ± std, 1k samples) for different methods and step counts. Note: DDPM EM at 1000 steps is the baseline; "—" means the method is not applicable.

Steps	Flow Matching	DDIM	DDPM EM
1	[result]	[result]	—
5	[result]	[result]	—
10	[result]	[result]	—
50	[result]	[result]	—
100	[result]	[result]	—
200	[result]	[result]	—
1000	[result]	[result]	*baseline*

7.2 Problem 6.B: Euler Sampler and Full Comparison

1. Flow matching typically produces recognizable FashionMNIST samples at around 5–10 steps, while DDPM EM may need 50–100 steps.
2. Flow matching needs approximately 10–50 steps to match 1000-step DDPM EM quality.
3. At 100 steps, flow matching and DDIM are both competitive with 1000-step DDPM EM. If they are comparable, the bottleneck is the model capacity, not the sampler.
4. For best quality: DDPM EM at 1000 steps. For fast generation: Flow Matching at 10–50 steps.

7.3 Problem 6.C: One-Step Generation after Reflow

Table 2: Reflow comparison (1-step generation).

Method	Steps	Sample grid	FID	Time (s/64)
Rect. Flow — Reflow	1	[see figure]	[result]	[result]

One-step reflow dramatically outperforms one-step rectified flow without reflow, because the reflow procedure straightens the ODE trajectories, making a single Euler step accurate. Compared to 1000-step DDPM, one-step reflow trades some quality for 1000× faster generation.

7.4 Problem 6.D: Side-by-Side Qualitative Grid

The same initial noise vector tends to produce semantically similar samples across methods when many steps are used (DDPM EM, RF 100 steps), as the denoising trajectory converges to the same attractor basin. With fewer steps (1-step RF, 1-step reflow), the correspondence is weaker for the non-reflow model but stronger for the reflowed model, since reflow makes the velocity field close to a linear map, preserving the global structure of the noise.

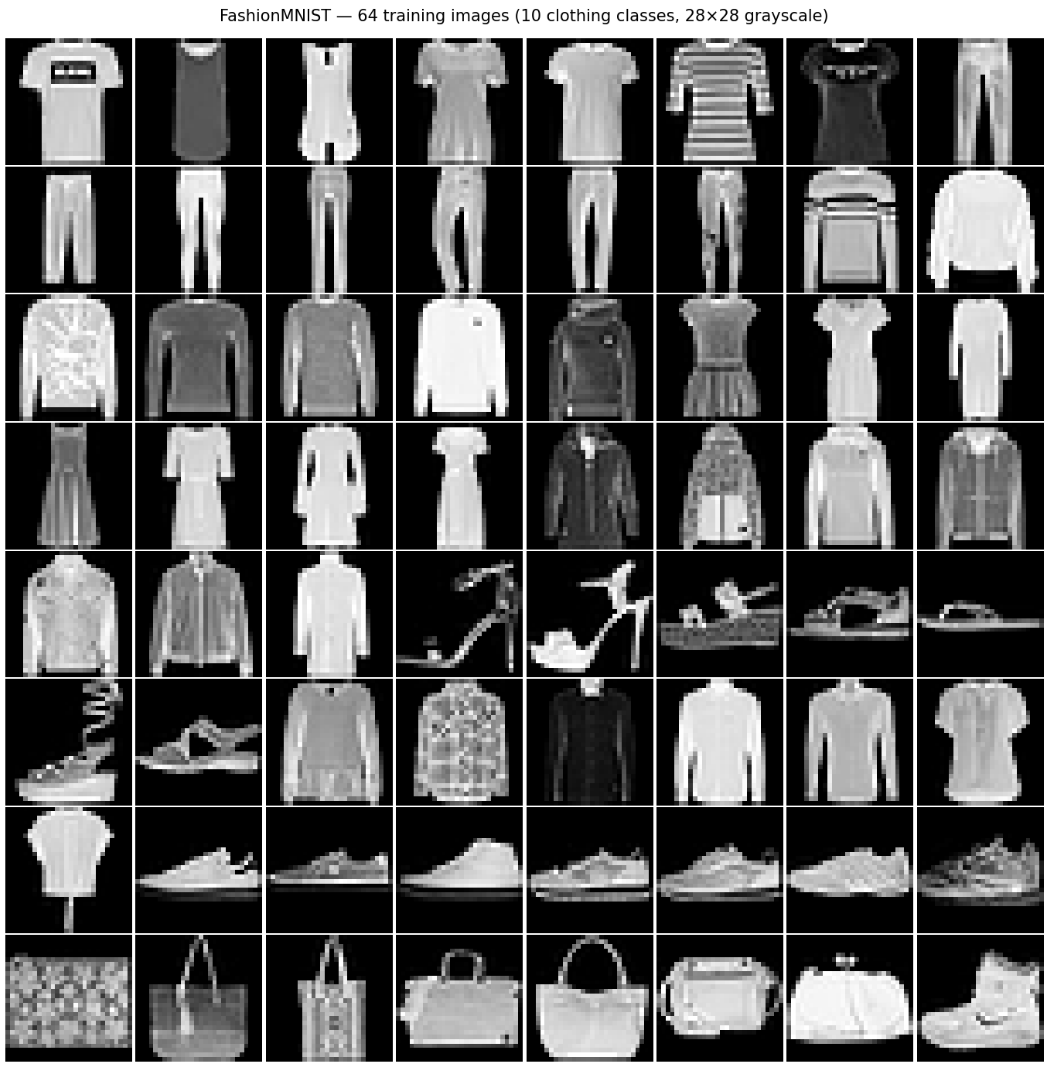

Figure 2: 64 FashionMNIST training images arranged in an 8×8 grid. Each image is 28×28 grayscale. The dataset spans 10 clothing classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot.

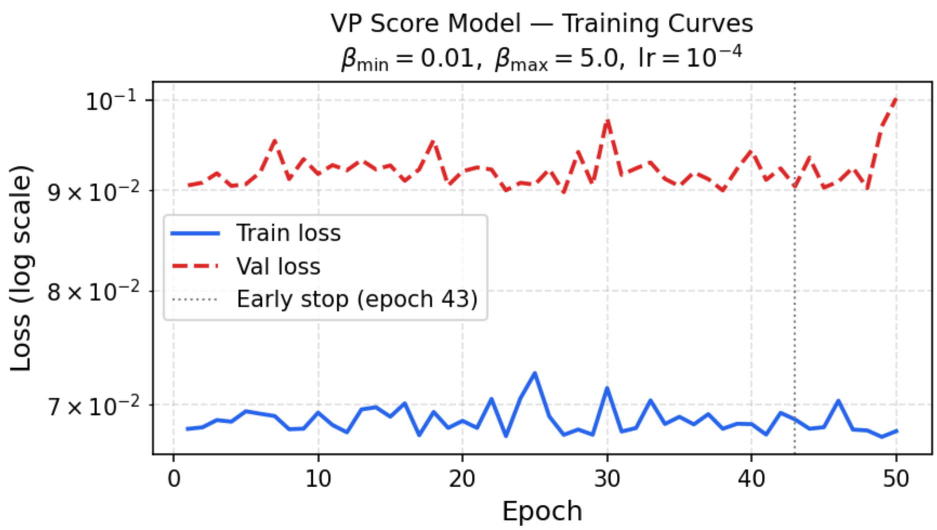

Figure 3: VP score model training and validation loss (log scale) over 50 epochs. $\beta_{\min} = 0.01$, $\beta_{\max} = 5.0$, lr$= 10^{-4}$, batch size$= 128$. Early stopping with patience$= 10$ was applied (stopped at epoch 43).

Figure 4: 64 samples generated by the Euler-Maruyama reverse-SDE sampler (1000 steps) from the trained VP score model.

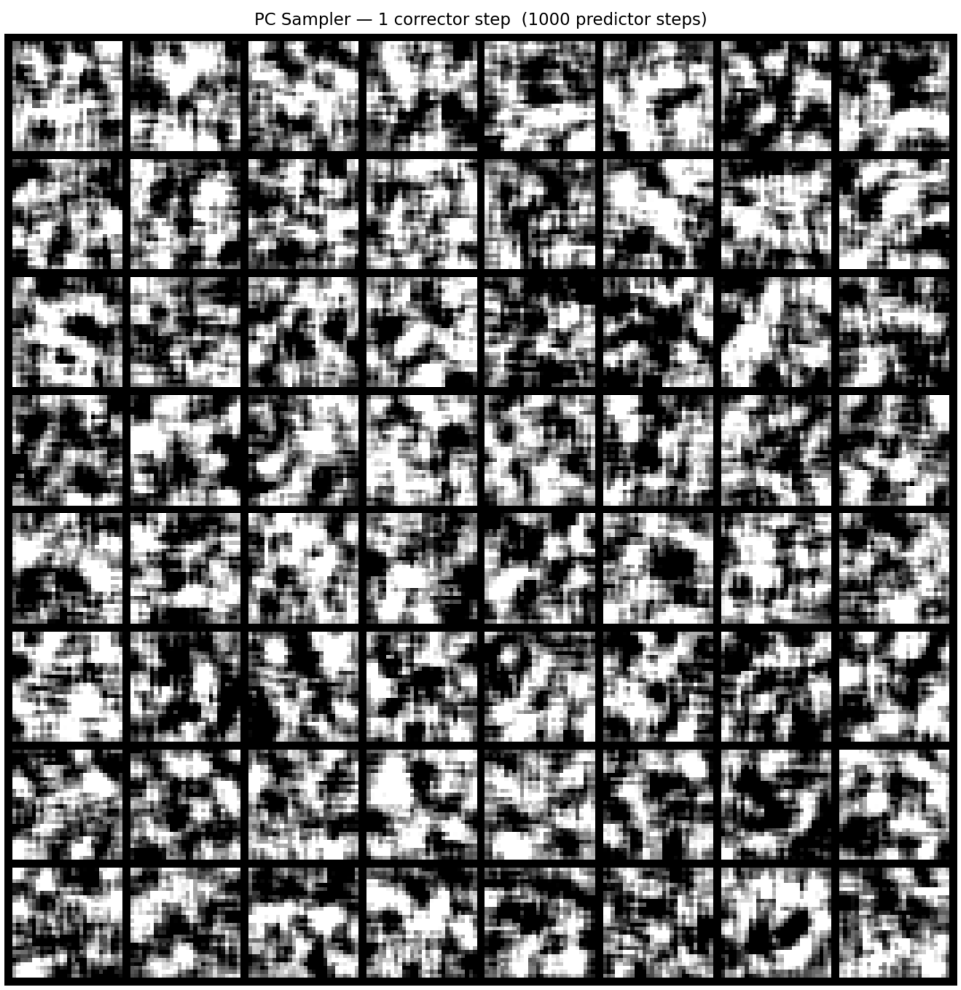

Figure 5: PC sampler — 1 corrector step per predictor step (1000 predictor steps).

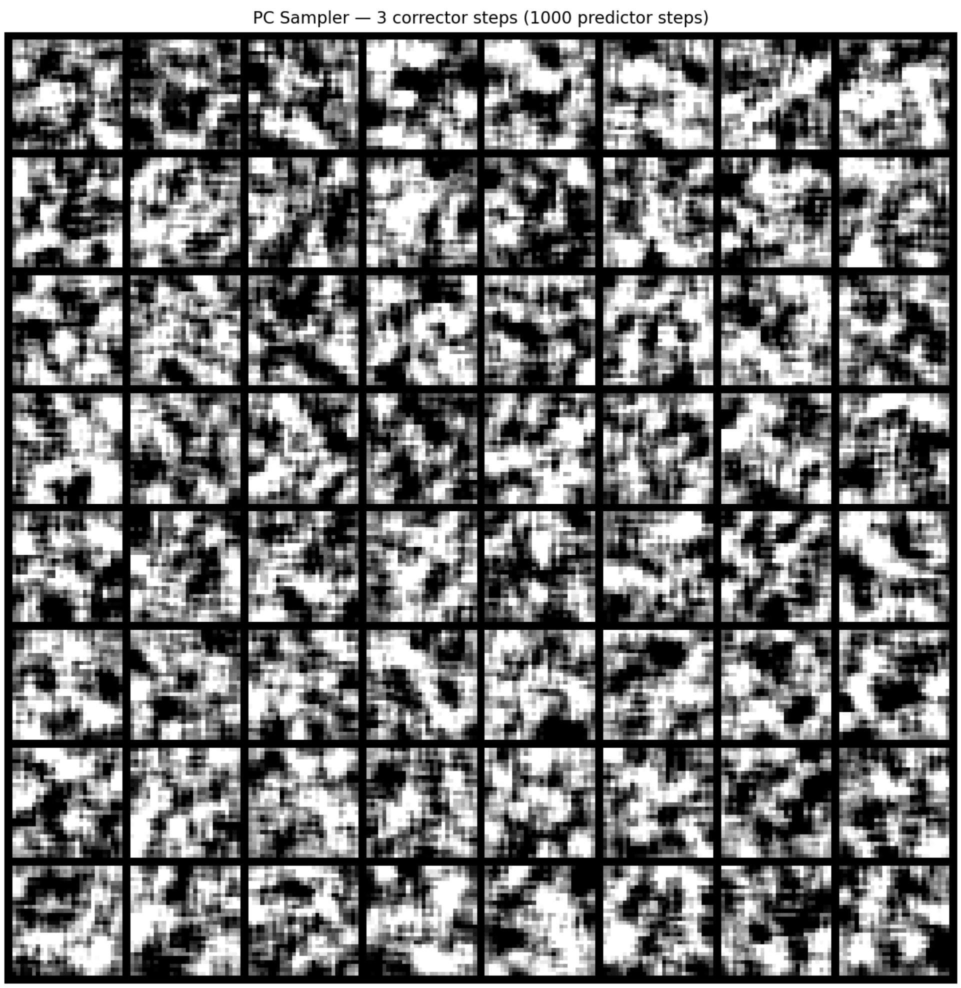

Figure 6: PC sampler — 3 corrector steps per predictor step (1000 predictor steps). Additional corrector steps reduce accumulated discretization error, producing slightly sharper and more coherent samples.

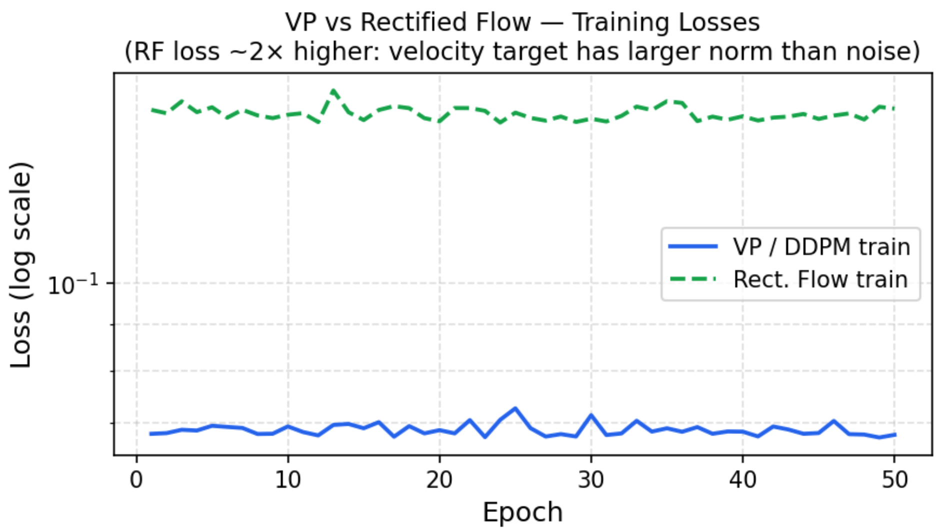

Figure 7: VP (DDPM) and Rectified Flow training losses plotted on the same log-scale axes. The RF loss is approximately 2× higher because the velocity target $X_1 - X_0$ has a larger norm than the unit-Gaussian noise target ε.

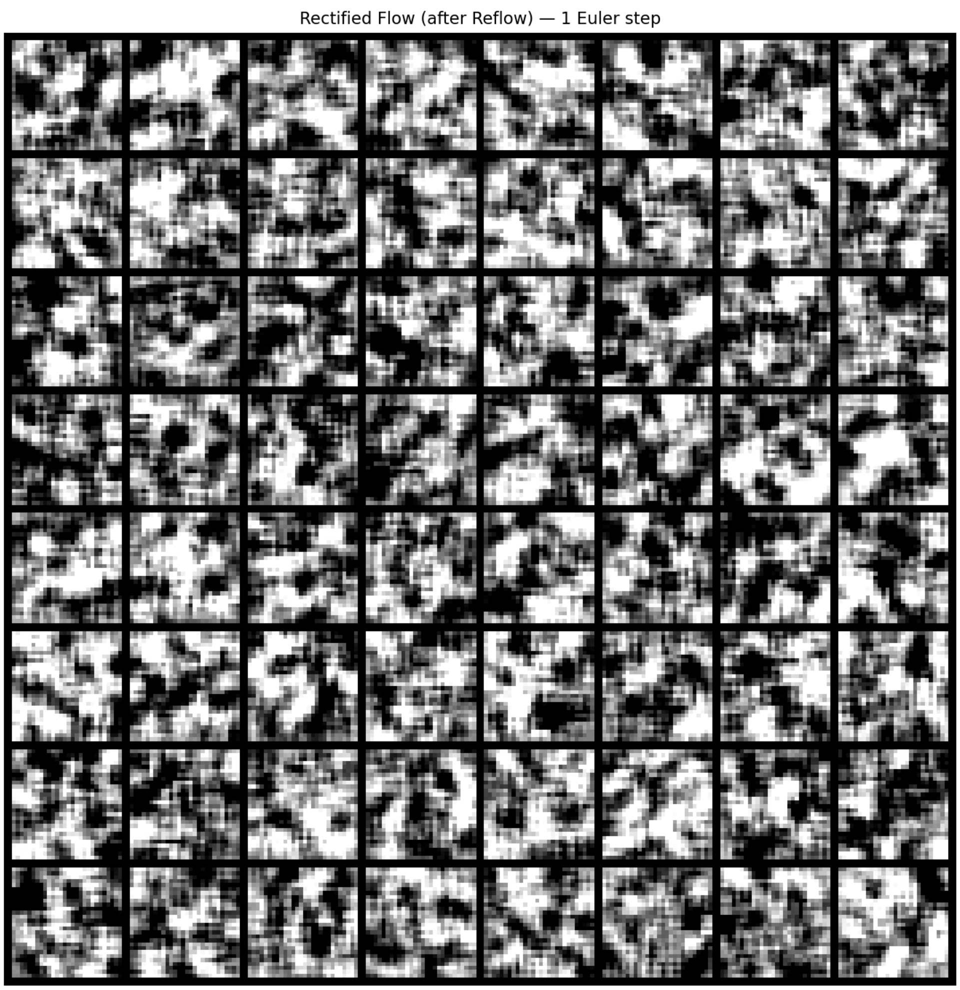

Figure 8: Rectified Flow (after one round of reflow) — 64 samples with a *single* Euler step. Straightened trajectories allow one-step generation that is visibly sharper than one-step generation without reflow.

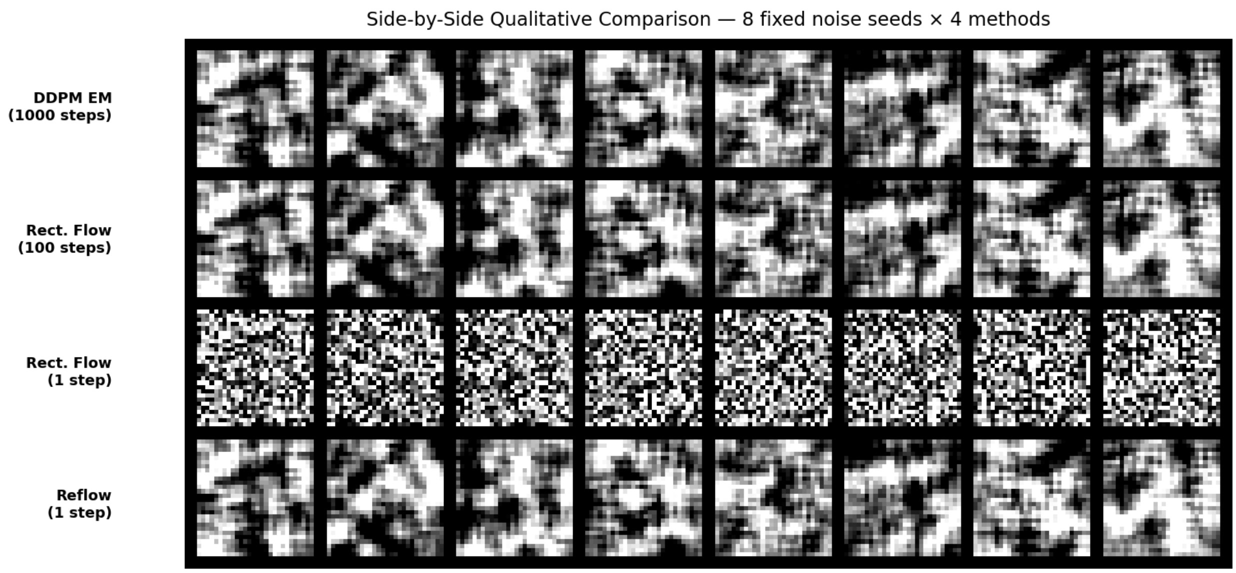

Figure 9: Side-by-side qualitative comparison using 8 fixed initial noise seeds. Each column is one seed; each row is one method. Row 1: DDPM EM (1000 steps). Row 2: Rect. Flow (100 steps). Row 3: Rect. Flow (1 step, no reflow). Row 4: Reflow (1 step, after one round of reflow).